Yewon Choi

Ph.D. Candidate, Integrative Medicine, Yonsei University College of Medicine, Seoul, Korea
Ph.D. expected **February 2027** (defense December 2026) · Available from **March 2027**
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Research Interest

I build trustworthy and auditable medical AI: benchmarks, metrics and LLM agents that show not only what a model decided but whether its reasons match the documented evidence. My current focus is evidence synthesis. I independently designed and built a citation-screening benchmark with reviewer-documented exclusion reasons (CsPICO), a reason-attribution metric (RA-F1), and an evidence-guided retrieval method for LLM screening (cKATE). I also fine-tune compact vision-language models on clinician-corrected rationales and design multi-agent LLM systems for clinical documentation under hospital IRB approval.

- Auditable LLM agents for systematic reviews and evidence synthesis
- Evaluation of clinical LLM and VLM reasoning (reason attribution, rationale quality)
- Clinical prediction and validation on large-scale EHR, cohort and imaging data

Education

Yonsei University College of Medicine, Seoul, Korea Mar 2022 – Feb 2027 (expected)
Integrated M.S.–Ph.D., Department of Integrative Medicine (Digital Healthcare) ·
Advisor: Prof. Sungjun Kim
Dissertation: Use-Aligned, Verifiable, and Auditable Medical AI Systems (defense scheduled December 2026)

CHA University, Korea Mar 2018 – Feb 2022
B.S. in AI Healthcare; double major in Data Science & Management
Project: Regulatory Approval of AI-Based Medical Devices: Current Status and Policy Implications

Selected Research

CsPICO: auditable LLM-assisted citation screening 2025 – present
Sole first author; independently designed and implemented the benchmark, metric and method

- Built **CsPICO**, a benchmark of 458 systematic reviews and 6,795 full-text candidate papers with reviewer-documented, PICOS-structured exclusion reasons, plus a 2025 temporal hold-out set.
- Introduced **RA-F1**, which scores whether a model attributes an exclusion to the correct PICOS criterion; it exposes right-decision, wrong-reason errors that screening F1 cannot detect.
- Designed and implemented **cKATE**, a training-free method that retrieves the closest met and violated examples from PICOS evidence spans; it improved both metrics in tested settings (e.g., screening F1 76.1 → 78.4 with Gemma on TEST; RA-F1 52.7 → 57.0 with Qwen on EXT25).
- Evaluated 12 LLMs × 4 prompting strategies with an automated LangGraph/LangChain pipeline (embedding retrieval, Pydantic structured outputs). EMNLP 2026 DocInsights Workshop · OpenReview · code and data

PISR: rationale-grounded pressure-injury staging with a compact VLM 2026
Co-first author; dataset, training and evaluation design

- Curated 208 pressure-injury photographs with staging labels and rationales corrected by two wound-care nurses; built 98 preference pairs (nurse correction preferred over the AI draft).
- Fine-tuned MedGemma-4B with LoRA SFT followed by ORPO: 81.8% staging accuracy vs 70.0% for Gemini 3.1 Pro Preview, with under-staging of 1.8% vs 20.0%; reproduced ConceptVLM and FT-ARM and compared DPO and SimPO.
- Proposed **PIRAS**, a four-axis rationale-quality score: +0.119 over a stage-only SFT model at the same accuracy (0.681 vs 0.562), close to Gemini (0.695).

Multi-agent LLM system for nursing documentation 2025 – present

Working-level project lead, national R&D project (Ministry of Trade, Industry & Energy, 2025–2029)

- Designed a multi-agent architecture (document understanding, NER and SNOMED CT/FHIR code mapping, drafting, verification, summarization) for 10 nursing-record types, with rule-based and single-LLM baselines.
- Defined the evaluation plan with clinician-reviewed references (nurses and physicians with 5+ years of experience) and wrote the Severance Hospital IRB protocol and case report form (inpatient EMR, Jan 2023 – Nov 2025, de-identified, analysed in the hospital's secure environment).

CDP: cross-country divergence in LLM-simulated survey populations 2026

Co-first author; metric, generation pipeline, experiments and writing

- Defined cultural divergence (CD) and its preservation relative to human surveys (CDP); generated 300 synthetic respondents per country and setting with 4 LLMs × 3 persona prompts (World Values Survey, 6 countries; Big Five, 3 countries).
- Showed that the prompt with the lowest Wasserstein distance in 7 of 8 settings flattened cross-country differences the most in all 8 (CDP 14–59%). EMNLP 2026 PANDORA Workshop · arXiv:2609.29928

PUBLICATIONS

* equal contribution; † corresponding author

Peer-reviewed journal articles

A Novel Radiomics-based Interpretable Model for Bladder Cancer Grade Prediction Using White-Light Cystoscopy Images

Yewon Choi, Sang Wouk Cho, Junho Hong, Jongsoo Lee, Hwiyoung Kim†, Kwang Suk Lee†
European Urology Open Science, 87, 71–79 (2026). IF 4.5

Liver Cancer Risk Stratification Using Deep Learning on Nationwide Longitudinal Health Screening Data: A Retrospective Cohort Study

Yewon Choi, Sungmin Cho, Changdai Gu, Chungho Kim, Bomi Park†, Hwiyoung Kim†
BMC Medical Informatics and Decision Making, 26:44 (2026). IF 5.5

Alcohol amount, drinking patterns, and gastric cancer risk: a nationwide study using targeted maximum likelihood estimation

Chungho Kim, Byungmi Kim, Hye Ah Lee, Yewon Choi, Hwiyoung Kim†, Bomi Park†
European Journal of Epidemiology, 1–14 (2026). IF 6.6

An Analysis of Healthcare Application Research and Standardization Trends on 3D Body Scanner

Yewon Choi, Jisu Hong, Jungwon Kim, Jongsoo Han†, Young Huh†
Journal of the Society for Standards Certification and Safety, 13(4), 1–18 (2023). KCI

EMNLP 2026 workshop papers

CsPICO: A Unified Benchmark for Citation Screening in Systematic Reviews beyond Classification to Explain Exclusion Reasons

Yewon Choi, Hyun Do Jung, Dain Ji, Hwiyoung Kim†, Sungjun Kim†, Jaehyung Kim†
EMNLP 2026 Workshop on DocInsights, Budapest (poster, to be presented Oct 2026) · OpenReview

Cultural Divergence Preservation: Diagnosing Flattening and Caricature in LLM-Simulated Survey Populations

Yeeun Chae*, Yewon Choi*, Seunghyun Lee, Il Im†
EMNLP 2026 Workshop PANDORA, Budapest (to be presented Oct 2026) · arXiv:2609.29928

Manuscripts under review

PISR: A Benchmark for Rationale-Based Pressure Injury Staging and a Lightweight Vision-Language Model
Yewon Choi*, Junyoung Lee*, Dongyoung Yoon*, Yeonju Kim, Taeyoon Park, Mona Choi†, Hwiyoung Kim†
Under review

Predictive Performance and Reasoning Alignment of General-Purpose Large Language Models for Early Mortality Risk Stratification in Intensive Care: Multicohort Retrospective Evaluation Study
Sungmin Cho*, Soojung Choi*, Yewon Choi, et al., Hyungchul Lee†, Hwiyoung Kim†
Under review

CONFERENCE ABSTRACTS

International

A Visually Interpretable Radiomics Model for Bladder Cancer Grading Using White-Light Cystoscopy Images
Yewon Choi, Sang Wouk Cho, Junho Hong, Jongsoo Lee, Hwiyoung Kim†, Kwang Suk Lee† · EAU Congress 2026, European Urology abstract (2026)

A Comparative Study of the Sagittal and Coronal Plane Balance of the Spine Measured Using a 3D Full Body Scanner and Spine X-Ray
Tae Hoon Kang, Minseok Choi, Yewon Choi, Jungwon Kim, Jae Hyup Lee†, Yohan Lee†, Minjoon Cho† · 3DBODY.TECH 2023

Domestic (Korea)

PISR: A Benchmark for Rationale-Based Pressure Injury Staging and a Lightweight VLM
Yewon Choi*, Junyoung Lee*, Dongyoung Yoon*, et al., Mona Choi†, Hwiyoung Kim† · Korean Society of Artificial Intelligence in Medicine (KOSAIM) 2026 (poster, to be presented)

Developing an Early Mortality Prediction Model for ICU Patients Using LLM-Based Approaches
Soojung Choi, Sungmin Cho, Yewon Choi, Minsun Kim, Hwiyoung Kim · KOSAIM 2025 · Best Poster Presentation Award (LLM & Agent Track)

Liver Cancer Risk Stratification via Deep Learning on Longitudinal Nationwide Health Screening Data
Yewon Choi, Sungmin Cho, Changdai Gu, Chungho Kim, Bomi Park†, Hwiyoung Kim† · KOSAIM 2025

A Study on Standardization Trends for 3D Full-Body Scanner Utilization in Healthcare
Yewon Choi, Jungwon Kim, Jisu Hong · ICS 2023 Symposium on Information and Control

WORK EXPERIENCE

Severance Hospital Mar 2025 – present
AI Researcher, Institute for Innovation in Digital Healthcare (IIDH)
Lead the LLM-based nursing-documentation project (multi-agent design, evaluation plan, IRB protocol); developed the pressure-injury VLM and the citation-screening pipeline above.

Yonsei University College of Medicine, Medical AI Lab Mar 2022 – present
Graduate Researcher
Working-level lead and project manager for three national R&D projects (administration, budget, research and reporting); first-author studies on liver-cancer risk prediction and bladder-cancer radiomics.

VATA Co., Ltd. Apr 2024 – Mar 2026
Co-Founder & Co-CEO
Built and launched Clonefit (iOS/Android), which estimates a 3D body avatar and body composition from 2D photos; ran it for one year with about 100 users. Led planning, algorithm design and fundraising: KRW 120 million (≈ US$87,000) from Naver D2SF, the Ministry of SMEs and Startups, and Yonsei University Graduate School of AI.

Korea AI Medical Healthcare Research Institute (KoAIM) Aug 2021 – Apr 2023
Researcher, R&D Planning
Developed 3D body-scan AI models for chronic-disease management; managed a 3D body-shape analysis R&D project and consulted on standardization of AI-based medical devices.

Gangnam CHA Hospital, Infection Control Office Jan 2020 – Mar 2020
Intern

Research Projects

Working-level lead and project manager
- Development of LLM-based techniques and platforms for nursing-record generation and task automation (MOTIE/KEIT, RS-2025-02220286) (Apr 2025 – Dec 2029)
- AI-based prediction of cancer high-risk groups (Korea Health Industry Development Institute) (May 2023 – Dec 2024)
- 3D human body-shape analysis device (Korea Evaluation Institute of Industrial Technology) (Aug 2021 – May 2023)

Technical Skills

Programming	Python, R, SQL; PyTorch; Git, Docker, Weights & Biases
LLMs and agents	LangGraph, LangChain, Pydantic structured outputs, embedding retrieval (FAISS), RAG, LLM-as-a-judge evaluation
Model training	LoRA, supervised fine-tuning, preference optimization (ORPO, DPO, SimPO), CNN/MLP, gradient boosting
Clinical data	EHR/EMR, nationwide cohorts (NHIS), medical imaging and radiomics, SNOMED CT/FHIR, IRB protocols, de-identification
Methods	Systematic-review methods (PICOS, screening, exclusion reasons), survival analysis (Cox), SHAP, external validation; causal inference (TMLE, co-author)
Languages	Korean (native), English (OPIc IH, 2026)

Patents & Books

Apparatus, Method, and Computer-Readable Medium for the Visual Explanation of Bladder Cancer Grade Prediction
Kwang Suk Lee, Sang Wouk Cho, Jongsoo Lee, Hwiyoung Kim, Yewon Choi
Korean patent application 10-2025-0130194 (2025)

Everything About Medical Imaging AI, Part I: From Development Environment Setup to Machine-Learning Model Development
Sang Wouk Cho, Yewon Choi, Junho Hong, Hyundo Jung
WikiDocs e-book (2025) · wikidocs.net/book/10449

Honors & Awards

Best Poster Presentation Award, LLM & Agent Track, KOSAIM 2025
"Developing an Early Mortality Prediction Model for ICU Patients Using LLM-Based Approaches"

7th Place, AOCR 2024 AI Challenge (second stage) 2024
Kaggle competition on acute appendicitis classification from contrast-enhanced CT

Teaching

Teaching Assistant, Yonsei University Graduate School
- Medical Image Processing: Understanding and Applications (MRA7003) Spring 2024
- Introduction to Artificial Intelligence in Medicine (MRA7008) Fall 2023
- Topics in Medical Radiation Physics (MRA7002) Spring 2023

References

Available upon request.